# Linhao Bai

Robotics Software | Robot Learning | Sim-to-Real | Research Engineering

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## Experience

**Tesla** *Austin, TX*

Robotics Integrations Engineer Sep 2026 – Present

- Integrate **production AMR and ASRS software** across fleet orchestration, task sequencing, localization, robot runtime / firmware, and **PLC / OPC UA** interfaces to factory equipment.
- Develop and debug multi-robot **resource constraints, recovery workflows, and station handoffs**; validate physical execution using robot telemetry, Python tooling, and distributed-service logs.

**Georgia Institute of Technology – STAR Lab** *Atlanta, GA*

Graduate Research Assistant, Robot Learning Oct 2024 – May 2026

- Built a **ROS learned-policy-to-hardware control interface** for a Kinova arm and PSYONIC / Allegro dexterous hands, integrating robot SDKs, joint-state feedback, nonblocking velocity commands, arm/hand synchronization, timeouts, and experiment logging.
- Modified and evaluated **TD-MPC / TD-MPC2 world-model dynamics** in model-based RL experiments; compared MLP, linear, and **SINDy-regularized** variants using predictive error and planning returns, exposing dynamics–planner mismatch.
- Extended **Diffusion Policy training and evaluation** for DexArt manipulation: object-flow observation datasets, custom training workspace, checkpoint loading, and offline action-error evaluation across tasks. Code
- Used **MuJoCo, NVIDIA Isaac Sim / Isaac Gym** for robot-learning baselines, custom experiments, simulated-manipulation evaluation, and **sim-to-real** development; contributed to Video2Sim2Real perception, residual RL, and real-robot execution.

**SEER Robotics** *Shanghai, China*

Project Manager / Software Engineer Jun 2022 – Jul 2024

- Developed **C++ robot-controller services and integration plugins** for OTA / backup, time synchronization, **health monitoring**, and telemetry; debugged firmware, LiDAR / localization, and fleet-to-robot interfaces.
- Delivered AMR solutions to **35+ international clients**; led a **5-AMR / 100+ rack** warehouse system with SLAM + QR localization correction, rack reservations, and randomized multi-robot stress tests across hundreds of moves.

## Selected Robotics Research

- **Going with the Flow: Koopman Behavioral Models as Pseudo Planners for Visuo-Motor Dexterity** – *CoRL 2026, equal contribution.* Learned visual-flow / Koopman latent dynamics for long-horizon prediction and event-triggered replanning. Project | Paper
- **Video2Sim2Real: Full-Stack Autonomous Dexterous Skill Acquisition from a Single Human Video** – *2026 preprint.* Video / scene reconstruction → simulation / residual RL → physical manipulation. Project | Paper

## Education

**Georgia Institute of Technology** – M.S. Computer Science, **GPA 4.0 / 4.0** *May 2026*

**Chongqing Jiaotong University** – B.Eng. Civil Engineering *Jun 2022*

## Technical Skills

**Programming / Systems:** Python, C++, Bash, Linux, Git, Docker, Kubernetes, logging / telemetry, debugging

**Robotics / Hardware:** ROS / ROS2, Kinova and dexterous-hand SDKs, actuator control, sensor feedback, CAN, RS-485, OPC UA, LiDAR, SLAM, calibration

**Simulation / ML:** MuJoCo, NVIDIA Isaac Sim, Isaac Gym, PyTorch, TD-MPC, MPPI, SINDy, Koopman dynamics, Diffusion Policy, imitation / reinforcement learning, sim-to-real